Mean Value

Questions 1–21

Question 1. Find the mean value of each of the following on the given intervals:

a) $f(x) = 5x - 2$ on the interval $[3, 8]$
b) $f(x) = 5x^2$ on the interval $[2, 5]$
c) $f(x) = \frac{1}{\sqrt{x}}$ on the interval $[9, 16]$

Question 2. Find the mean value of the function

$$f(x) = 2x^2 - 1$$

in the interval $-0.5 \leq x \leq 1.2$, giving your answer to 3 significant figures.

Question 3. Show that the mean value of

$$f(x) = x^2$$

on the interval $[0, k]$ where $k > 0$ is $\frac{1}{3}k^2$

Question 4. Find the mean value of the function

$$f(x) = 8x^3$$

in the interval $0 \leq x \leq a$, giving your answer in terms of a

Question 5. Show that the mean value of

$$f(x) = mx + c$$

on the interval $[a, b]$ where m, c, a and b are constants and $a < b$ is

$$\frac{1}{2}m(a + b) + c$$

Question 6. Find the mean value of the function

$$f(x) = (x + 4)^2$$

in the interval $a \leq x \leq 2a$, giving your answer in the form $pa^2 + qa + r$ where p, q and r are constants to be found.

Question 7. Find the mean value of the function

$$f(x) = \frac{1+x}{\sqrt{x}}$$

in the interval $0 \leq x \leq k$, giving your answer in the form

$$\frac{p}{\sqrt{k}} + \frac{p\sqrt{k}}{q}$$

where p and q are constants to be found.

Question 8. The mean value of the function

$$f(x) = 2\sqrt{x} + k$$

in the interval $[1, 9]$ where k is a positive constant is $\frac{22}{3}$
Find the value of k

Question 9. The mean value of the function

$$f(x) = x^2 + 1$$

in the interval $[2, 5]$ is half the mean value of $f(x)$ for the interval $[0, k]$ where k is a positive integer. Find the exact value of k

Question 10. The mean value of the function

$$f(x) = 3x^2 + 12x - 40$$

in the interval $[0, k]$ where k is a positive constant is 51
Find the exact value of k

Question 11. The mean value of the function

$$f(x) = ax - 4$$

is -5.5 over the interval $[-2, 1]$

a) Determine the value of a

It is known that $f(x)$ also has a mean value of 18.5 over the interval $[p, 2p]$

b) Determine the value of p

Question 12. The velocity of a particle after t seconds is given by

$$v = \frac{1}{6}\sqrt{t}\left(12 - \frac{1}{t}\right)$$

a) Show that the mean velocity for $1 \leq t \leq 3$ is 2.68 ms^{-1} to 3 significant figures.

b) Calculate the mean acceleration of the particle for the same time period giving your answer to 3 significant figures.

Question 13. The temperature of a metal rod within some machinery at a distance x metres from one end is modelled by $\mathrm{T}(x) = -5x^3 + 12x^2 + 4x$ where $\mathrm{T}(x)$ is the temperature in degrees Celsius and x is the distance in metres from one end of the rod. The rod is 3 metres long. Find the mean temperature of the rod over its entire length.

Question 14. The growth of a population of rabbits $\mathrm{P}(t)$ during the first 12 months of the year is modelled by $\mathrm{P}(t) = 2t^3 - 7t^2 + 15t + 40$ where $\mathrm{P}(t)$ represents the number of rabbits and t is the number of months that have passed. Find the average population of rabbits over the first 12 months.

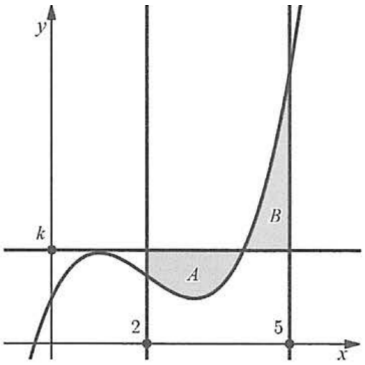
Question 15.

The diagram below shows the curve

$$f(x) = x^3 - 6x^2 + 9x + 4$$

with shaded regions A and B enclosed by the lines

$$y = k, x = 2 \text{ and } x = 5$$



Given that the areas A and B are equal, find the value of k

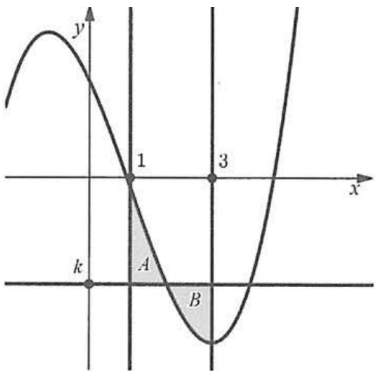
Question 16.

The diagram below shows the curve

$$f(x) = x^3 - 3x^2 - 9x + 10$$

with shaded regions A and B enclosed by the lines

$$y = k, x = 1 \text{ and } x = 3$$



Given that the areas A and B are equal, find the value of k

Question 17. A function

$$f(x) = x^3 - 2$$

has a mean value of

$$\frac{195}{4}$$

over the interval $[2, k]$ where k is a positive constant.
Find the exact value of k

Question 18. Given that the function f is defined by

$$f(x) = x^2 + x - 1 \text{ for } 2 \leq x \leq 4$$

a) Find the mean value of f

Another function g is defined by

$$g(x) = f(x) + c \text{ for } 2 \leq x \leq 4$$

b) Given that the mean value of g is 25, find the value of c

Question 19. A function f has mean value k over the interval

$$0 \leq x \leq 4$$

where k is an unknown constant.
In each case, find the mean value over the given interval in terms of k

a) $g(x)$ which is a translation of $y = f(x)$ by $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ over the interval $-2 \leq x \leq 2$
b) $h(x)$ which is a reflection in the x-axis of $y = f(x)$ over the interval $0 \leq x \leq 4$
c) $m(x)$ which is a reflection in the y-axis of $y = f(x)$ over the interval $-4 \leq x \leq 0$
d) $n(x)$ which is a translation of $y = m(x)$ by $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ over the interval $-3 \leq x \leq 1$

Question 20. A function f has mean value k over the interval

$$2 \leq x \leq 7$$

where k is an unknown constant.
In each case, find the mean value over the given interval in terms of k

a) $g(x)$ which is a translation of $y = f(x)$ by $\begin{pmatrix} -1 \\ 4 \end{pmatrix}$ over the interval $1 \leq x \leq 6$
b) $h(x)$ which is a stretch, parallel to the x-axis, scale factor 3 of $y = f(x)$ over the interval $6 \leq x \leq 21$
c) $m(x)$ which is a stretch, parallel to the y-axis, scale factor $\frac{1}{2}$ of $y = f(x)$ over the interval $2 \leq x \leq 7$

Question 21. The root mean square of a function $f(x)$ defined for $a \leq x \leq b$ over this interval is

$$\sqrt{\frac{1}{b-a}\int_a^b [f(x)]^2 \, \mathrm{d}x}$$

Find the root mean square of

$$f(x) = -x(x-0.5)(x-1)$$

from $x = 0.1$ to $x = 0.6$ giving your answer to 3 significant figures.